

Department of Applied Mathematics & Computational Science
B.E. I Year (Sec A to E) (Jan.-June 2017)
Mid Term Test I
MA-10501: Mathematics I

Time: 1 hour

Max. Marks :15

Note : Attempt any three . All questions carry equal marks.

Q.1	Reduce the matrix to normal form and find its rank	$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$	(05)
Q.2	Find the inverse of a matrix by Gauss -Jordan Method	$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$	(05)
Q.3	Solve:	$(x - y - 2)dx + (x - 2y - 3)dy = 0$	(05)
Q.4	Solve:	$(1 + y^2)dx + (x - e^{-\tan^{-1}y})dy = 0$	(05)

Department of Applied Mathematics & Computational Science
Mid-Term Test-II (July-Dec. 2016)
B.E. I Year
MA 10501: Mathematics- II

Time: 01 Hour

Max. Marks: 20

Note: Attempt any four questions. All questions carry equal marks.

Q.1	Solve: $(1 + 2x)^2 \frac{d^2y}{dx^2} - 6(1+2x) \frac{dy}{dx} + 16y = 8(1+2x)^2$	(05)
Q.2	Solve: $\frac{dx}{dt} + 2 \frac{dy}{dt} - 2x + 2y = 3e^t$, $3 \frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 4e^{2t}$	(05)
Q.3	If $x = \cos \theta + i \sin \theta$, $y = \cos \phi + i \sin \phi$ then prove that: $\frac{x-y}{x+y} = i \tan \left(\frac{\theta-\phi}{2} \right)$	(05)
Q.4	Find all Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	(05)
Q.5	If 10 % of bolts produced by a machine are defective. Determine the probability that out of 10 bolts, chosen at random i. One ii. None iii. At most 2 bolts will be defective.	(05)

Department of Applied Mathematics & Computational Science
Mid-Term Test-I (July-Dec. 2015)
B.E. I Year MA 1056 : Mathematics- II

Time: 01 Hour

Max. Marks: 20

Note: Attempt any **four** questions. All questions carry equal marks.

Q.1	Solve $\frac{dy}{dx} = \frac{x^2+xy}{x^2+y^2}$	(05)
Q.2(i)	Solve $6y^2 dx - x(2x^3 + y)dy = 0$	2.5+2.5
(ii)	Find the inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & 2 & 1 \end{bmatrix}$	
Q.3	Find non-singular matrices P and Q such that PAQ is in the normal form of A where $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$	(05)
Q.4	Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$	(05)
Q.5	Show that three equations $-2x + y + z = a$ $x - 2y + z = b$ $x + y - 2z = c$ have no solution unless $a + b + c = 0$ in which case they have infinitely solutions find the solutions when $a = 1, b = 1, c = -2$	(05)

Department of Applied Mathematics & Computational Science
Mid-Term Test-I (July 2016-Dec.2016)
B.E. I Year (Section F to J)
MA 10501 : Mathematics- II

Time: 01 Hour

Max. Marks: 15

Note: Attempt any **three** questions. All questions carry equal marks.

Q.1	Solve $(xy^2 - e^{1/x^3})dx - x^2y dy = 0$	(05)
Q.2	Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2x + e^{2x}$	(05)
Q.3	Find the rank of the matrix: $\begin{bmatrix} 2 & 3 & -2 & 4 \\ 3 & -2 & -1 & 2 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{bmatrix}$	(05)
Q.4	For what values of k the system of equations: $x + y + z = 1$ $2x + y + 4z = k$ $4x + y + 10z = k^2$ has a solution.	(05)

BE I Year
Mid Term Test II (Dec. '14-April'15)
Mathematics II

Time: 1 Hrs.]

[Max. Marks : 20]

Note: Attempt any four questions. All questions carry equal marks.

- Q.1 Solve $(x^3 D^3 - 3x^2 D^2 + x D) y = x + \log(x)$ (05)
- Q.2 Solve $(D^2 - 1)y = x \sin(x) + x^2 e^x$ (05)
- Q.3 100 toys are found to be defective in a lot of 5000 toys. Find the probability that atmost 3 toys are defective in a box of 100 toys. [$e^{-2} = 0.1353$] (05)
- Q.4 In a sample of 1000 cases, the mean of a certain test is 14 and S.D. is 2.5. Assuming the distribution to be normal find:
(i) How much students score between 12 and 15?
(ii) How many score above 16?
(iii) How many score below 8? (05)
- Q.5 Fit a straight line to the following data treating y as dependent variable (05)
- | | | | | | |
|---|---|---|---|----|----|
| x | 1 | 2 | 3 | 4 | 5 |
| y | 2 | 7 | 9 | 10 | 11 |

Mid-Term Test I (Jan.-June 2015)
B.E. I Year
Mathematics II: MA 1056

[Max. Marks: 20]

Time: 1 hour]

Note : Attempt any four questions. All questions carry equal marks.

Q.1	Solve. $(x \sec^2 y - x^2 \cos y) dy = (\tan y - 3x^4) dx$	(05)
Q.2(i)	Solve. $x \log x \frac{dy}{dx} + y = 2 \log x$.	(2.5+2.5)
(ii)	Find the nature, index and signature of the given quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_1x_3$	
Q.3	Find the rank of the following matrices by reducing it into normal form $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	(05)
Q.4	Find the eigen values and eigen vectors of the following matrix $\begin{bmatrix} 6 & 2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	(05)
Q.5	Solve. $\frac{dy}{dx} = 2y \tan x + y^2 \tan^2 x$.	(05)

$$\frac{dy}{dx} + y f(x) = (1 - y) e^{4x/2}$$

Department of Applied Mathematics & Computational Science
Mid-Term Test-I (July-Dec. 2015)
B.E. I Year MA 1056 : Mathematics- II

Time: 01 Hour

Note: Attempt any four questions. All questions carry equal marks.

Max. Marks: 20

Q.1	Solve $\frac{dy}{dx} = \frac{x^2+xy}{x^2+y^2}$	(05)
Q.2(i)	Solve $6y^2 dx - x(2x^3 + y)dy = 0$	2.5+2.5
(ii)	Find the inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$	
Q.3	Find non-singular matrices P and Q such that PAQ is in the normal form of A where $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$	(05)
Q.4	Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$	(05)
Q.5	Show that three equations $-2x + y + z = a$ $x - 2y + z = b$ $x + y - 2z = c$ have no solution unless $a + b + c = 0$ in which case they have infinitely solutions find these solutions when $a = 1, b = 1, c = -2$	(05)

Department of Applied Mathematics & Computational Science
Mid-Term Test-I (July-Dec. 2015)
B.E. I Year MA 1056 : Mathematics- II

Time: 01 Hour

Note: Attempt any four questions. All questions carry equal marks.

Max. Marks: 20

Q.1	Solve $\frac{dy}{dx} = \frac{x^2+xy}{x^2+y^2}$	(05)
Q.2(i)	Solve $6y^2 dx - x(2x^3 + y)dy = 0$	2.5+2.5
(ii)	Find the inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ $\frac{dy}{dx}$	
Q.3	Find non-singular matrices P and Q such that PAQ is in the normal form of A where $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ $1 + 3xy^2$ $4xy^2 + 2$	(05)
Q.4	Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$	(05)
Q.5	Show that three equations $-2x + y + z = a$ $x - 2y + z = b$ $x + y - 2z = c$ have no solution unless $a + b + c = 0$ in which case they have infinitely solutions find these solutions when $a = 1, b = 1, c = -2$ xy^{k+1} $y(x^{k+1})$	(05)

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1(i)	Find the inverse of A where $\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$	(03)
(ii)	Solve the following differential equation: $\frac{dy}{dx} = \frac{y+x-2}{y-x-4}$	(02)
Q.2	Reduce the Matrix A to its Normal form, where $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ & hence find the Rank of the Matrix.	(05)
Q.3	Solve completely the following system of equations: $\begin{aligned} 2w + 3x - y - z &= 0 \\ 4w - 6x - 2y + 2z &= 0 \\ -6w + 12x + 3y - 4z &= 0 \end{aligned}$	(05)
Q.4	Solve $\frac{dy}{dx} = \frac{y+1}{(y+2)e^x - x}$	(05)
Q.5	Solve $\{ 2xy \cos x^2 - 2xy + 1 \} dx + \{ \sin x^2 - x^2 + 3 \} dy = 0$	(05)

Time : 1hr.]

[Max.Marks : 20

Note : Attempt any FOUR questions.

Q.1 Reduce the Matrix A to the normal form,

$$A = \begin{bmatrix} 3 & -4 & -1 & 2 \\ 1 & 7 & 3 & 1 \\ 5 & -2 & 5 & 4 \\ 9 & -3 & 7 & 7 \end{bmatrix}$$

Hence find the rank of A

Rank - 2

(05)

Q.2 If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 9 & 3 \\ 1 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, Verify that

(05)

$$(AB)^{-1} = B^{-1}A^{-1}$$

Q.3 Find the rank of matrices

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(05)

Q.4 Solve the differential equation

(i) $y' + 4xy + xy^3 = 0$

(05)

(ii) $(y^2x + 2x^2y^3)dx + (x^2y - x^3y^2)dy = 0$

Q.5 Solve. $(xy^2 - x^2)dx + (3x^2y^2 + x^2y - 2x^3 + y^2)dy = 0$

(05)

6
-
4
5

Time : 1hr.]

[Max.Marks : 20

Note : Attempt any FOUR questions.

Q.1 Reduce the Matrix A to the normal form ,

$$A = \begin{bmatrix} 3 & -4 & -1 & 2 \\ 1 & 7 & 3 & 1 \\ 5 & -2 & 5 & 4 \\ 9 & -3 & 7 & 7 \end{bmatrix}$$

Rank - 2

(05)

Hence find the rank of A

Q.2 If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 9 & 3 \\ 1 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, Verify that

(05)

$$(AB)^{-1} = B^{-1} \cdot A^{-1}$$

Q.3 Find the rank of matrices

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(05)

Q.4 Solve the differential equation

(05)

(i) $y' + 4xy + xy^3 = 0$

(ii) $(y^2x + 2x^2y^3)dx + (x^2y - x^3y^2)dy = 0$

Q.5 Solve. $(xy^2 - x^2)dx + (3x^2y^2 + x^2y - 2x^3 + y^2)dy = 0$

(05)

-1+2

~~Ret B + 12~~

8-6:4

Department of Applied Mathematics & Computational Science

B.E. I Year (July-Nov. 2013)

Mid Term Test II

MA 1056 : Mathematics II Sec F, G, H, I

Time: 1 Hour

Note: Attempt any Four Questions.

Max. Marks

Q.1 Find all the Eigen values and Eigen vectors of the matrix. (05)

$$A = \begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Q.2 Solve:- $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^x + \cos 2x + x^2$ (05)

Q.3 Solve:- $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$ (05)

Q.4 Using matrix method, show that the equation
 $3x + 3y + 2z = 1$; $x + 2y = 4$; $10y + 3z = -2$; $2x - 3y - z = 5$
 are consistent and hence obtain the solution for x,y,z. (05)

Q.5 Solve by method of variation of parameter.
 $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^3 e^x$ (05)

B.E. I Year (July-Nov. 2013)

Mid Term Test II

MA 1056 : Mathematics II Sec F, G, H, I

Time: 1 Hour

Note: Attempt any Four Questions.

Max. Marks: 20

Q.1 Find all the Eigen values and Eigen vectors of the matrix. (05)

$$A = \begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Q.2 Solve:- $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^x + \cos 2x + x^2$ (05)

Q.3 Solve:- $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$ (05)

Q.4 Using matrix method, show that the equation
 $3x + 3y + 2z = 1$; $x + 2y = 4$; $10y + 3z = -2$; $2x - 3y - z = 5$
 are consistent and hence obtain the solution for x,y,z. (05)

Q.5 Solve by method of variation of parameter. (05)

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^3 e^x$$

NOV-DEC 2013 EXAMINATION

I B.E. (4YDC) EXAM

MA-1056/MA-1054/A110/ MA 153/MA 155

MATHEMATICS II

Time: 3 Hrs.]

[Max. Marks : 70

[Min. Pass Marks : 22

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Attempt all the questions. All questions carry equal marks. Each question shall carry four subparts a, b, c and d. Choice should be given to attempt any one from a and b and similarly any one from c and d.

Q.1 (a) Find the rank of the matrix

MARKS

07

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

OR

(b) Using Matrix method, show that the equations

07

$$3x + 3y + 2z = 1$$

$$x + 2y = 4$$

$$10y + 3z = -2$$

$$2x - 3y - z = 5$$

are consistent and hence contain the solutions for x, y and z.

(c) Find the eigen values and eigen vectors of :

07

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

Determine whether the eigen vectors are orthogonal.

OR

(d) Reduce the quadratic form $x^2 - 4y^2 + 6z^2 + 2xy - 4xz + 2w^2 - 6zw$ into sum of squares.

07

Q.2 (a) Solve the following differential equations

$$(i) (3y + 2x + 4)dx - (4x + 6y + 5)dy = 0$$

07

$$(ii) (xy^2 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$$

(b) Find the complete solution of

OR

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = xe^{3x} + \sin 2x$$

07

(c) Solve the simultaneous equations

$$\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2\cos t - 7\sin t,$$

$$\frac{dx}{dt} - \frac{dy}{dt} + 2x = 4\cos t - 3\sin t$$

OR

07

Class Test -I (Mar.- 2013)
1AM051- Applied Mathematics -II
For (IT.& Civil)

Time :70 min.

Maximum Marks:25

Note : Attempt all questions.

Q1 Find the inverse of $A = \begin{bmatrix} 1 & 3/2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ by Partition method. *eigen value & eigen vector*

Q2. For what values of k the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ has a solution.

Q3. Find the Fourier series for the Function $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in $(0, 2\pi)$.

Q4. Expand the function $f(x) = x \sin x$ as a Fourier series in the interval $-\pi \leq x \leq \pi$.